

The cocyclic origin of the manifestly finite period polynomials and L -functions

Conceptual companion to `dr_b_periods_and_Lfunctions.tex`

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This note explains, at the level of *why* rather than *how*, what ties the period-polynomial construction of §§2–3 to the L -function of §4 in the main file. It is the conceptual skeleton; the verifications all live in the source. It is also a direct answer to the two margin notes in the source (at the cohomology/periods paragraph of §2 and the Γ -equivariance remark), which ask for exactly this tutorial.

The single thread. *The period polynomial and the L -function are not two constructions. They are one object — a single gauge-fixed cocycle generator value — read at two values of a complex parameter s : integer s gives the period polynomial, general s gives the L -function. The Hurwitz kernel $\tilde{k}_T(\tau, s)$ is the unique device that gauge-fixes that cocycle, and the same equation that defines it is, simultaneously, (i) the coboundary equation whose solvability is cuspidality, (ii) the identity that makes the two-segment functional finite and basepoint-independent, and (iii) the all- s continuation of the Bernoulli inversion that produces the classical period polynomial. “Manifestly finite” and “cocyclic” are the same statement.*

1 Why a cocycle at all?

The object one wants is the period polynomial, classically $r_f(X, Y) = \int_0^{i\infty} f(\tau)(X - \tau Y)^{k-2} d\tau$. The integral runs between two cusps, $0 = S \cdot i\infty$ and $i\infty$, so it is morally “the S -edge with the basepoint pushed to the cusp.” For a genuine cusp form this converges and one stops. For a *weakly holomorphic* or meromorphic form it diverges at the cusp, and the classical contour is unavailable.

The cocycle is the replacement for that broken open contour. Set $n := k - 2$, let $V_n = \mathbb{C}[X, Y]_n$ carry the weight- n slash action, and form the V_n -valued 1-form $\omega_f(\tau) = f(\tau)(X - \tau Y)^n d\tau$. The content of the Γ -equivariance remark (the first margin note) is exactly this: the weight- k transformation of f cancels the weight- $(-n)$ slash on $(X - \tau Y)^n$, so ω_f is invariant under the *simultaneous* action of Γ on τ and on (X, Y) . That is the one fact that makes everything downstream work, and it is the whole reason the auxiliary variables X, Y are introduced.

Equivariance lets us trade the single divergent open contour for a *system of finite contours, one per group element*:

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f \quad \in \quad V_n \otimes \mathbb{C}.$$

Path-additivity through the intermediate point $h^{-1}\tau_0$, together with equivariance, gives the cocycle relation

$$C_{gh}^f = C_g^f|_h + C_h^f.$$

This relation is the bookkeeping that “closes the contour under Γ ”: it says the γ -integrals are mutually consistent under composition, which is precisely the structure a boundary-at-infinity integral loses. A cocycle on $\mathrm{SL}_2(\mathbb{Z}) = \langle S, T \rangle$ is determined by its two generator values

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \int_{-1/\tau_0}^{\tau_0} \omega_f, \quad C_T^f(\tau_0) = (2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} \omega_f.$$

Both run between *finite* interior points of $\mathbb{H}$. Nothing has gone to the cusp; this is where “manifestly finite” begins.

What the cohomology paragraph means (the second margin note). Varying the base-point τ_0 changes C_γ^f by a coboundary $\delta P(\gamma) = P|_\gamma - P$ (provided no pole is crossed), so the class $[C^f] \in H^1(\Gamma; V_n)$ is basepoint-free. The Eichler–Shimura theorem says the cuspidal part of $H^1(\Gamma; V_n)$ is a copy of $S_k \oplus \overline{S_k}$, and the coordinates of $[C^f]$ in that space are the periods $\omega^\pm(f)$ and quasi-periods $\eta^\pm(f)$. In plain terms: *the period polynomial is a set of coordinates on a cohomology class, and a cohomology class is a cocycle modulo coboundaries*. So to read off the periods you must first pick a concrete cocycle representative — you must *gauge-fix* the coboundary freedom. That is the entire job of §3, and the next two sections are what the gauge-fixing costs.

2 Why cancel the T -variation?

Among the generators, T is special: in the presentation $\langle S, T \mid S^4 = (ST)^6 = 1 \rangle$ it is the one generator carrying *no* relation of its own. It is therefore the coboundary direction one is free to spend. Concretely, T acts on V_n as the translation $X \mapsto X + Y$, so $\delta_T := (\cdot)|_T - \mathrm{id} = e^{Y\partial_X} - 1$ is strictly nilpotent, with

$$\ker \delta_T = \mathbb{C} \cdot Y^n, \quad \mathrm{im} \delta_T = \{Q : [Q]_{X^n} = 0\}.$$

Now the decisive computation. The X^n -coefficient of C_T^f is the bare T -segment integral $\int_{\tau_0-1}^{\tau_0} f(\tau) d\tau$, and unit-segment T -periodicity collapses this to the constant Fourier coefficient:

$$[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} c_f(0).$$

Hence the coboundary equation

$$\boxed{\delta_T P = (2\pi i)^{-(n+1)} C_T^f(\tau_0)}$$

is solvable for $P \in V_n$ *if and only if* $c_f(0) = 0$, i.e. if and only if f is a (weak) cusp form. This is the conceptual punchline of §3: *cuspidality is not an analytic hypothesis here; it is exactly the solvability of the equation that kills the T -variation*. The constant mode $c_f(0)$ — the Eisenstein-direction obstruction — is precisely the single coefficient δ_T cannot reach, and a cusp form is one for which it vanishes.

So we cancel the T -variation because that is the gauge-fixing demanded by §2: removing C_T^f as a coboundary δP pushes the non-period (constant-mode / boundary-monomial) content out of the cohomologically-meaningful coordinates and leaves a representative that genuinely *is* the period polynomial. The leftover is the single generator value

$$r_f := C_S^f(\tau_0) - (2\pi i)^{n+1} (P|_S - P).$$

A subtlety worth stating precisely. r_f is *one* generator value — the S -value of the gauge-fixed cocycle $C' = C - \delta((2\pi i)^{n+1}P)$. It is *not* $C_S + C_T$. The period relations on r_f (the S - and U -relations) come from the group relations *conditioned on* $C_T = 0$: once T is gauged, $C_U = C_S|_T$, so the third generator is not independent data and the U -relation reappears as a constraint on C_S . This is why two numbers, not four, come out, and why it is wrong to think of the construction as adding two cycles.

3 Why does cancelling T introduce the kernel $\tilde{k}_T$?

This is the step that looks mysterious until one sees that the kernel is nothing but the coboundary correction, re-expressed as an integral.

The gauge-fixed value r_f carries the explicit subtraction $-(2\pi i)^{n+1}(P|_S - P)$. To turn that into something one can read, solve the boxed coboundary equation in closed form. Because $\delta_T = e^{Y\partial_X} - 1$ and $Y\partial_X$ is nilpotent on V_n , the Bernoulli generating function $z/(e^z - 1) = \sum_k B_k z^k/k!$ truncates and supplies the exact inverse

$$\delta_T^{-1} = \sum_{k \geq 0} \frac{B_k}{k!} (Y\partial_X)^{k-1}.$$

Feeding the T -moments of C_T^f through this inverse and collecting the matched monomial coefficient, the coboundary subtraction $-(P|_S - P)$ *re-expresses itself as a T -segment integral against a $\mathbb{Q}$ -rational polynomial kernel* $\hat{k}_T(\ell, \tau)$ — equivalently, after a constant shift that integrates to zero on a cusp form, against the Hurwitz polynomial $\tilde{k}_T(\tau, \ell)$. The upshot, for each interior ℓ , is

$$[r_f]_{X^{n-(\ell-1)}Y^{\ell-1}} = (2\pi i)^{n+1}(-1)^{\ell-1} \binom{n}{\ell-1} \left[\underbrace{\int_{\gamma_{\tau_0}^S} f \tau^{\ell-1} d\tau}_{\text{bare } S\text{-piece}} + \underbrace{\int_{\gamma_{\tau_0}^T} f \tilde{k}_T(\tau, \ell) d\tau}_{\text{the coboundary, as a } T\text{-integral}} \right].$$

So the “ T -kernel” is not a second physical contribution. It is the gauge-fixing coboundary $-(P|_S - P)$ wearing the costume of a T -integral. This is the honest content of the two-segment formula: it is one gauge-fixed generator value, and the second term is what the coboundary subtraction looks like once δ_T^{-1} is written out.

Why Hurwitz, specifically. Drop the integer/polynomial restriction and ask for the same gauge-fixing at a general power τ^{s-1} . One needs a kernel solving the unit-shift equation

$$\tilde{k}_T(\tau, s) - \tilde{k}_T(\tau - 1, s) = -\tau^{s-1} + e^{i\pi(s-1)} \tau^{k-1-s}.$$

The function that inverts a unit shift against a single power is the Hurwitz zeta, $\zeta(\sigma, \tau+1) - \zeta(\sigma, \tau) = -\tau^{-\sigma}$, so a combination of two Hurwitz zetas,

$$\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1),$$

is the natural solution. The Bernoulli inversion of §3 is exactly its integer- s shadow, through $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ — Bernoulli numbers being zeta values at non-positive integers. The two are not analogous; they are the same inversion at two ranges of s .

4 The unification: “manifestly finite” is “cocyclic”

Now the thread closes. The shift-difference equation of the previous section is not a coincidence of the kernel’s definition — it is forced by basepoint-independence. Differentiating the two-segment functional

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = i^{-s} \left[\int_{\gamma_{\tau_0}^S} f \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f \tilde{k}_T(\tau, s) d\tau \right]$$

in τ_0 , only endpoint terms survive, and the modularity relations $f(\tau_0 - 1) = f(\tau_0)$, $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ collapse the τ_0 -derivative to exactly

$$i^{-s} f(\tau_0) [\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) + \tau_0^{s-1} - e^{i\pi(s-1)} \tau_0^{k-1-s}] = 0.$$

The bracket vanishes *by the shift-difference identity*. Read this slowly: τ_0 -independence of L^* holds *if and only if* the kernel satisfies the shift-difference equation, which at integer s is the coboundary equation $\delta_T P = C_T^f$ that gauge-fixes the cocycle. Basepoint-independence of the L -function and the killability of the T -cocycle are one statement.

This gives the whole picture in one line:

$L\text{-function (general } s) \xleftarrow{\text{same gauge-fixed value}} \text{period polynomial (integer } s),$ both finite and basepoint-free <i>because</i> the kernel solves the coboundary / shift-difference equation.

And the finiteness is manifest, not engineered:

- The two contours run between finite interior points of $\mathbb{H}$. The divergent cusp limit of the classical Mellin contour is simply never taken — it has been replaced by the cocycle’s closed-under- Γ bookkeeping. There is no regulator, no asymptotic subtraction, no analytic continuation in the definition.
- What *would* have been the divergent boundary term at the cusp is instead the finite coboundary correction $-(P|_S - P)$, living on a compact T -segment and represented by $\tilde{k}_T$.
- For weakly holomorphic f (no interior poles) $\mathbb{H}$ is simply connected, so local constancy upgrades to global τ_0 -independence: a single well-defined $L^*(f, s)$. Evaluating at the elliptic fixed point $\tau_0 = i$ collapses the S -segment to a point, leaving one integral on $[i-1, i]$ whose unfolding is the completed L -function. The BFK non-holomorphic regulator is then not a separate definition but *one closed-form evaluation* of this finite functional at $\tau_0 = i$; that is why the two agree term by term, and why L^* is independent of any regularisation choice.
- For meromorphic f , simple connectivity fails, local constancy is all one can have, and the residue picked up on crossing an interior pole is the wall-crossing term. The same functional, now winding-dependent, is the meromorphic L -function.

5 What is and is not being claimed

To keep the signal clean: the unification above is an organising *reading* of the construction in the source, not an independent theorem. Its three legs — cuspidality $\Leftrightarrow \delta_T$ - solvability, τ_0 -independence $\Leftrightarrow$ the shift-difference identity, and Bernoulli-as-integer-shadow-of-Hurwitz — are each established

in the file (Lemmas on δ_T , the local-constancy lemma, and the Hurwitz-collapse lemma respectively). What this note adds is only the observation that they are the *same* equation seen three ways, which is what makes “manifestly finite” and “cocyclic” synonymous here.

The genuinely load-bearing identification — that the finite-contour functional and the BFK regularised value are equal, not merely structurally similar — rests on the term-by-term unfolding at $\tau_0 = i$ and on the normalisation matching between the two constructions. That is a computation, not a slogan, and it is the right place to keep one’s guard up: agreement “up to an elementary $\Gamma/2\pi/i^k$ factor” is a weaker statement than identity, and the honest check is the symbol-for-symbol comparison against the BFK values at a fixed weight. The conceptual story does not substitute for that check; it only explains why the two objects are candidates to coincide.